DO NOT WRITE IN THIS AREA

12 Prove algebraically that

$$(m + 3)^2 - (m - 3)^2 + 3 \quad \text{is odd}$$

for all integer values of m.

$(m + 3)^2 - (m - 3)^2 + 3$

1 mark
Expand both squares.

$= m^2 + 6m + 9 - (m^2 - 6m + 9) + 3$

$= 12m + 3$

1 mark
Distribute the minus over the whole bracket.

$= 2(6m + 1) + 1$

6m + 1 is an integer, so the expression is odd.

1 mark
An odd integer has the form 2k + 1.

(Total for Question 12 is 3 marks)